Day 077 | Unit 06 | Teacher plan

Understand and Use Ratios and Rates

Generate Equivalent Ratios

Standards: 6.RP.A.3a

Learning target: I can complete an equivalent-ratio table.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Multiply both values in a ratio by the same scale factor. Organize pairs in a table and check that every row compares the same quantities in the same order.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

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01 / Learn

Multiply both values in a ratio by the same scale factor. Organize pairs in a table and check that every row compares the same quantities in the same order.

Worked example 1

A mix uses 2 cups of oats for 6 cups of seeds. Complete the table (oats, seeds): (2, 6), (4, ___), (8, ___).

1. Multiply both values in a ratio by the same scale factor.

2. Organize pairs in a table and check that every row compares the same quantities in the same order.

3. Multiply both parts by 2 and by 4.

4. State the result in the form requested: 12; 24.

Answer: 12; 24

Worked example 2

A recipe uses 2 cups oats for 3 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (4, 6)
B. (9, 12)
C. (6, 9)
D. (4, 7)

1. Multiply both values in a ratio by the same scale factor.

2. Equivalent ratios multiply both quantities by the same factor.

3. A uses 2

4. B uses 3.

5. C and D change only one scaled quantity.

Answer: A, C - (4, 6); (6, 9)

Day 077 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A mix uses 8 cups of oats for 7 cups of seeds. Complete the table (oats, seeds): (8, 7), (16, ___), (16, ___).

02 / Constructed response

A mix uses 3 cups of oats for 3 cups of seeds. Complete the table (oats, seeds): (3, 3), (6, ___), (18, ___).

03 / Constructed response

A mix uses 5 cups of oats for 3 cups of seeds. Complete the table (oats, seeds): (5, 3), (10, ___), (10, ___).

04 / Constructed response

A mix uses 3 cups of oats for 4 cups of seeds. Complete the table (oats, seeds): (3, 4), (6, ___), (15, ___).

05 / Constructed response

A mix uses 8 cups of oats for 2 cups of seeds. Complete the table (oats, seeds): (8, 2), (16, ___), (16, ___).

06 / Constructed response

A mix uses 4 cups of oats for 3 cups of seeds. Complete the table (oats, seeds): (4, 3), (8, ___), (24, ___).

07 / Constructed response

A mix uses 7 cups of oats for 3 cups of seeds. Complete the table (oats, seeds): (7, 3), (14, ___), (28, ___) and explain how you know the last two rows are equivalent.

08 / Multiple select

A recipe uses 5 cups oats for 3 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (21, 12)
B. (15, 9)
C. (10, 7)
D. (10, 6)

09 / Multiple select

A recipe uses 2 cups oats for 2 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (4, 5)
B. (6, 6)
C. (4, 4)
D. (11, 10)

10 / Constructed response

A mix uses 6 cups of oats for 5 cups of seeds. Complete the table (oats, seeds): (6, 5), (12, ___), (42, ___) and explain how you know the last two rows are equivalent.

Day 077 | Unit 06 | Answer key

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1. 14; 14

Multiply both parts by 2 and by 2.

6.RP.A.3a | Constructed response

2. 6; 18

Multiply both parts by 2 and by 6.

6.RP.A.3a | Constructed response

3. 6; 6

Multiply both parts by 2 and by 2.

6.RP.A.3a | Constructed response

4. 8; 20

Multiply both parts by 2 and by 5.

6.RP.A.3a | Constructed response

5. 4; 4

Multiply both parts by 2 and by 2.

6.RP.A.3a | Constructed response

6. 6; 18

Multiply both parts by 2 and by 6.

6.RP.A.3a | Constructed response

7. 6; 12

Multiply both parts by 2 and by 4.

6.RP.A.3a | Constructed response

8. B, D - (15, 9); (10, 6)

Equivalent ratios multiply both quantities by the same factor. A uses 2; B uses 3. C and D change only one scaled quantity.

6.RP.A.3a | Multiple select

9. B, C - (6, 6); (4, 4)

Equivalent ratios multiply both quantities by the same factor. A uses 2; B uses 3. C and D change only one scaled quantity.

6.RP.A.3a | Multiple select

10. 10; 35

Multiply both parts by 2 and by 7.

6.RP.A.3a | Constructed response